

Project Report: Retail Data Analysis (Superstore Project)

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Tools Used: SQL Server , Power BI Desktop, Drowning.oi, Get Hub

1. Dataset Description

Dataset Components:

Customer Info: Names, Segments (Consumer, Corporate, Home Office), and geographic locations.

Order Info: Order IDs, Order Dates, Ship Dates, and Shipping Modes.

Product Info: Categories (Technology, Furniture, Office Supplies) and Sub-Categories.

Financial Metrics: Sales, Quantity, Discounts, and Profit.

Note: The data was so clear and clean, I needed just to change profit data type to money .

2. Database Architecture & SQL Development

This section covers the structural design and the technical implementation using SQL Server to manage the project data.

A. Relational Schema & Design Decisions

The database was transformed from a flat structure into a Relational Model to ensure data integrity. The core architecture includes:

Key Tables: Sales_Team (Staff details), Sample-Superstore (Sales data), Employee_Bonuses (Performance tracking), and Dependents (HR records).

Relationships: A One-to-Many (1:N) relationship was established between



the Sales_Team and the Superstore dataset using the Region column as a common link.

Constraints: We enforced Primary Keys for unique identification and Foreign Keys to maintain referential integrity between tables.

B. SQL Technical Implementation

Advanced SQL techniques were applied to clean, move, and analyze the data:

Automation: Used the IDENTITY(1,1) property to automate ID generation for new records.

Data Analysis: Written complex Joins and Subqueries to map sales performance to specific employees and identify high-value orders.

Logic Optimization: Developed Stored Procedures to automate repetitive tasks, such as retrieving individual employee performance reports with a single command.

3. Power BI Dashboard Development

The reporting phase focused on creating an interactive environment for stakeholders to monitor KPIs.

A. DAX Measures (The Analytical Core)

Three sophisticated DAX measures were created to handle time-intelligence and growth tracking:

- Total Sales & Profit: Basic aggregations using SUM().
- Sales YTD (Year To Date): * Logic: Calculates cumulative sales from the start of the current year.

DAX: TOTALYTD(SUM(Sales), 'Order Date')

- Sales Growth %: * Logic: Compares current performance against



the same period last year.

Formula: (Current Sales - Previous Year Sales) / Previous Year Sales. This involved handling "Division by Zero" errors using the DIVIDE function for stability.

DAX: Sales Growth % = DIVIDE([Total Sales] - [Sales PY], [Sales PY], 0)

B. Visualizations & Interactivity

The dashboard consists of 4 primary visuals designed for "Cross-Filtering":

KPI Cards: Highlighting Total Sales and Profit Margin % at a glance.

Bar Chart: Breakdown of Sales by Category to identify top-performing product lines.

Line Chart: Visualizing sales trends over time to detect seasonality.

Slicers: Interactive filters for Region and Segment, allowing managers to drill down into specific market pockets.

4. Strategic Insights & Findings

After analyzing the SQL outputs and Power BI visuals, the following insights were discovered:

Profitability vs. Volume: The Technology category generates the highest profit margins, even if it doesn't have the highest volume of orders.

Regional Performance: The West Region consistently hits YTD targets, while the Central Region shows a decline in growth due to high discount rates.

Shipping Efficiency: First-class shipping correlates with higher customer satisfaction but significantly reduces profit margins per order.

